

Genome-wide analysis of the *CCCH* zinc finger gene family in *Medicago truncatula*

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Abstract

Key message In this study, we identified 34 *CCCH* Znf genes in *Medicago truncatula* and the results of semi-quantitative RT-PCR revealed that the expression patterns of subfamily VI members were diverse.

Abstract *CCCH*-type zinc finger (Znf) proteins are specific transcriptional factors with a typical motif consisting of three cysteine residues and one histidine residue. Increasing evidences have revealed that *CCCH* Znf proteins participated in the regulation of plant growth, developmental processes and environmental responses. Survey and characterization of *CCCH* Znf genes in leguminous species would facilitate a better understanding of the evolutionary processes and functions of this gene family. In this study, we performed a comprehensive analysis of *CCCH* Znf genes in *M. truncatula* by describing the phylogenetic relationships, chromosomal location and gene structure of each family member. A total of 34 *CCCH* Znf genes were identified in the latest *M. truncatula* genome sequence. The 34 predicted members were clustered into nine subfamilies based on their phylogenetic analysis and

structure features. In addition, the 34 *Medicago CCCH* Znf genes were found to be unevenly distributed on eight chromosomes. Furthermore, the expression profiles of subfamily VI were investigated under different stress conditions (PEG-6000, NaCl and ABA) by using semi-quantitative RT-PCR. The data showed that these genes displayed different expression levels in response to various stress conditions. The results presented in this study provide basic information about *Medicago CCCH* Znf genes and form a fundamental clue for cloning genes with specific functions in further studies and applications.

Keywords *CCCH* zinc finger · Phylogenetic analysis · Gene structure · Expression profile · *Medicago truncatula*

Introduction

Transcription factors (TFs) are critical regulators of gene expressions that control many important biological processes, such as cellular morphogenesis, signal transduction and environmental stress responses (Riechmann et al. 2000; Wray et al. 2003). Zinc finger (Znf) transcription factors, as one of the largest TF families, are characterized by the common Znf motifs in which cysteines and/or histidines coordinate with a few zinc ions to form the local peptide structures that are essential for their specific functions (Hall 2005). Based on their structural characteristics and functional diversities, Znf transcription factors can be categorized into at least 14 families, such as *RING*-finger, *WRKY*, *DOF* and *LIM* families which have been identified in plants (Arnaud et al. 2007; Kosarev et al. 2002; Yanagisawa 2004; Zhang and Wang 2005). These gene families have been proved to regulate gene expression with the aid of DNA-binding or protein-binding proteins.

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However, (Wang et al. 2008a) discovered a new type of *Arabidopsis* zinc finger proteins, named *CCCH* zinc finger family, differing from the previously identified plant Znf TFs by regulating gene expression through directly binding to mRNA.

The *CCCH* zinc finger family contains a typical *CCCH*-type zinc finger motif and the members of this family had already been found in eukaryotes ranging from humans to yeast (Carrick et al. 2004; De et al. 1999; Thompson et al. 1996). In mouse, tristetraprolin (TTP), a well-characterized *CCCH* Znf protein containing two *CCCH* Znf motifs, binds to AU-rich elements within the 3'-untranslated region of their target transcripts to mediate mRNA degradation in most cases (Lai et al. 1999, 2000, 2003). Like its better-known relative TTP, Zfp3612 is an mRNA-binding and destabilizing protein and functions in the physiological control of female fertility during early embryonic development (Ramos et al. 2004). Another *CCCH*-type Znf protein, zinc finger antiviral protein (ZAP) isolated from Rat2 fibroblasts, can directly bind to specific viral RNA sequences through its *CCCH* Znf motifs and inhibit retroviral RNA production (Gao et al. 2002). The other two *CCCH* Znf proteins of *C. elegans*, *PIE-1* and *POS-1*, can both regulate germ cell fate by inhibition of transcription or activation of protein expression from maternal RNAs (Mello et al. 1996; Ogura et al. 2003; Tenenhaus et al. 2001).

Compared to the largely well-characterized *CCCH* Znf proteins of mammals, only a small number of *CCCH* Znf proteins have been functionally characterized in plants at present. These proteins have been implicated in participation in many plant developmental and adaptive processes, including seed germination (Kim et al. 2008), embryo development (Grabowska et al. 2009; Wang et al. 2008b), floral morphogenesis (Li et al. 2001), leaf senescence (Li and Thomas 1998) and plant architecture determination (Kong et al. 2006). In *Arabidopsis*, HUA1, a *CCCH*-type Znf protein with six tandem *CCCH* motifs, has been confirmed to be an RNA-binding protein and likely participates in a new regulatory mechanism governing flower development (Li et al. 2001). In addition, the other two *CCCH* Znf genes, *AtC3H14* (At1g66810) and *AtC3H15* (At1g68200), have recently been proved to regulate secondary cell wall biosynthesis in *Arabidopsis* (Ko et al. 2009; Wang et al. 2010), which also implies that their homologs may participate in *Arabidopsis* secondary cell wall formation as well. Presently, increasing evidences indicate a number of *CCCH* Znf genes involved in plant abiotic stresses and defense responses (Ko et al. 2009; Kong et al. 2006; Wang et al. 2008a, 2010). For example, two closely related proteins in *Arabidopsis*, *AtSZF1* (salt stress-inducible Znf protein 1) and *AtSZF2*, were both found to be involved in modulating the tolerance of plants

to salt stress (Sun et al. 2007). *GHZFP1*, a *CCCH*-type Znf protein from cotton, was demonstrated to enhance the tolerance to salt, drought, salicylic acid (SA) stress and fungal disease in transgenic tobacco (Guo et al. 2009). Recently, (Wang et al. 2008a) proved that 11 subfamily IX members of *Arabidopsis* *CCCH* Znf proteins participated in plant tolerance to different stresses such as drought, salt, cold and ABA.

Medicago truncatula is a diploid plant with a small genome, and its high genetic transformation efficiency makes it an excellent model organism for other leguminous plants. Although the characterizations of *CCCH* Znf proteins have been reported in some other species of plants and animals, their functions are poorly understood in *M. truncatula*. In this study, we performed a comprehensive analysis of 34 members of the *CCCH* Znf gene family in *M. truncatula*, including phylogenetic analysis, chromosomal location, gene structure and expression profiles. Our investigation would provide an important foundation for future cloning and functional studies of *CCCH* Znf proteins in *M. truncatula*.

Materials and methods

Identification of *CCCH* Znf proteins in *M. truncatula*

The latest version (V3.5) of the *M. truncatula* genome and protein sequences was downloaded from the *Medicago* genome database (<http://www.medicago.org/>). First, a local protein database of *M. truncatula* was constructed by using DNAtools software. Then the hidden Markov model (HMM) profiles of *CCCH* domains from the Pfam database (<http://pfam.sanger.ac.uk/>) were extracted from poplar, maize, *Arabidopsis* and rice (Chai et al. 2012; Peng et al. 2012; Wang et al. 2008a). The highly conserved sequences of *CCCH* Znf motifs (C-X₄₋₁₁-C-X₄₋₆-C-X₃-H) were used to search the *CCCH* Znf genes in *M. truncatula* protein database with the BLASTP program in the DNAtools software. The e-value for these searches was set as 1×10^{-3} . Finally, all obtained proteins were confirmed to be the real *CCCH* Znf proteins by Pfam and the confirmed *CCCH* Znf proteins were aligned using ClustalX to remove the redundant sequences. In addition, the three fields (length, molecular weight and isoelectric point) of each *Medicago* *CCCH* Znf protein were calculated by online ExPasy program (<http://www.expasy.org/tools/>).

Mapping *CCCH* Znf genes on chromosomes

The starting position of each *CCCH* Znf gene was obtained from the *Medicago* sequencing database. The locations of the 34 *Medicago* *CCCH* Znf genes were drafted by

MapInspect software (http://www.plantbreeding.wur.nl/uk/software_mapinspect.html). The candidate *CCCH* Znf genes were named from *MtC3H1* to *MtC3H34* according to their positions from the top to the bottom on the *Medicago* chromosomes (0–8).

Phylogenetic analysis and exon–intron structure of *CCCH* Znf genes

Multiple alignments of 34 *Medicago CCCH* amino acid sequences were performed with the ClustalX (version 1.83) (Thompson et al. 1997) and manually corrected. The phylogenetic trees were generated with ClustalX by using the neighbor-joining (NJ) method. Bootstrap values were calculated from 1,000 iterations.

The coding sequence (CDS) was obtained from the *Medicago* sequencing database. Exon–intron organizations of *CCCH* Znf genes were identified by comparing the coding sequences with their corresponding genomic sequences using GSDS software (<http://gsds.cbi.pku.edu.cn/>). To analyze gene duplication events, a multiple sequence alignment of the *Medicago CCCH* Znf proteins was performed by ClustalX and calculated using MEGA v4.0 (Tamura et al. 2007).

Plant material and treatments

M. truncatula cv. ‘Jemalog’ line A17 was used in this study. Plants were grown in the greenhouse at 25 ± 2 °C in the soil:sand mixture (3:1) and irrigated with water once every 2 days. The seedlings that had germinated after 6 weeks were selected for stress treatments according to Wang et al. (2008a). For drought stress and salt stress, the seedlings were treated with 300 mM PEG-6000 (polyethylene glycol) and 300 mM NaCl for 24 h, respectively. For ABA treatment, the seedling leaves were sprayed with 100 μ M ABA and sampled 30 min after treatment. Treatment with deionized water was performed as a control.

Expression profiling of *CCCH* Znf genes in *M. truncatula*

The genome-wide microarray data were obtained from the *M. truncatula* Gene Expression Atlas (MtGEA) Project website (<http://mtgea.noble.org/v2/>). The expression data were gene-wise normalized and hierarchically clustered based on the Pearson coefficients with average linkage in cluster 3.0 and Treeview software (<http://bonsai.hgc.jp/~mdehoon/software/cluster/>).

Total RNA was extracted from seedlings using the Trizol reagent (Bio Basic Inc, Canada). First-strand cDNA was synthesized using reverse transcriptase (TAKARA, Da Lian, China) according to the manufacturer’s instructions.

To investigate the expression patterns of *Medicago CCCH* Znf genes under different stresses, the subfamily VI genes were selected for analysis using semi-quantitative RT-PCR. Expression of the *Medicago* actin gene was used as an internal control. The gene-specific and control primers were designed using Primer Premier 5.0 software (Table S1). PCR reactions used Taq Polymerase (TAKARA, Da Lian, China) with the following profile: initial denaturation at 95 °C for 5 min, followed by 29 cycles (the actin gene was 25 cycles) of denaturation at 95 °C for 30 s, 30 s at 55 ± 2 °C, polymerization at 72 °C for 30 s and a final extension of 10 min at 72 °C. A 6 μ L sample of the PCR products was examined on a 1.0 % agarose gel with ethidium bromide.

Results

Characterization of *CCCH* Znf proteins in *M. truncatula*

To identify all members of the *CCCH* Znf family in *M. truncatula*, the HMM profiles of the *CCCH* Znf domains from *Arabidopsis*, rice, maize and poplar *CCCH* Znf genes were used to query the *CCCH* Znf genes in the protein database of *M. truncatula* using the BLASTP program. A total of 34 non-redundant *CCCH* Znf genes (named *MtC3H1*–*MtC3H34*) were confirmed in *M. truncatula*. The characterizations of the 34 *Medicago CCCH* Znf genes, including the number of amino acids (length), molecular weights, isoelectric points (PIs) and physical location, are listed in Table 1. According to detailed information, we found that the length of the 34 *CCCH* Znf proteins were variable: *MtC3H10* encodes a protein of 1,084 amino acids, while *MtC3H1* encodes only 139 amino acids. The isoelectric points of the 34 *CCCH* Znf genes ranged from 4.66 (*MtC3H30*) to 9.82 (*MtC3H32*).

In this study, 86 *CCCH* Znf motifs were identified by Pfam and SMART (Schultz et al. 2000; Sonnhammer et al. 1997) among the 34 predicted *Medicago CCCH* Znf proteins, which was fewer than those found in *Arabidopsis* (152) and rice (150) (Fig. 1a). Previous studies have indicated that the *CCCH* Znf proteins have between one and six copies of *CCCH*-type Znf motifs in animal or plant (Hudson et al. 2004; Kramer et al. 2010; Wang et al. 2008b). However, we identified a maximum of five copies of the *CCCH*-type Znf motifs in *M. truncatula*. Therefore, we divided the 34 *CCCH* Znf proteins into five groups according to the number of copies of *CCCH* Znf motif in each protein. As shown in Fig. 1b, 55.8 % of *Medicago CCCH* Znf proteins have one or two *CCCH* Znf motifs, while 23.5 % contain three copies and 14.7 % have five copies; these trends were also observed in *Arabidopsis* and

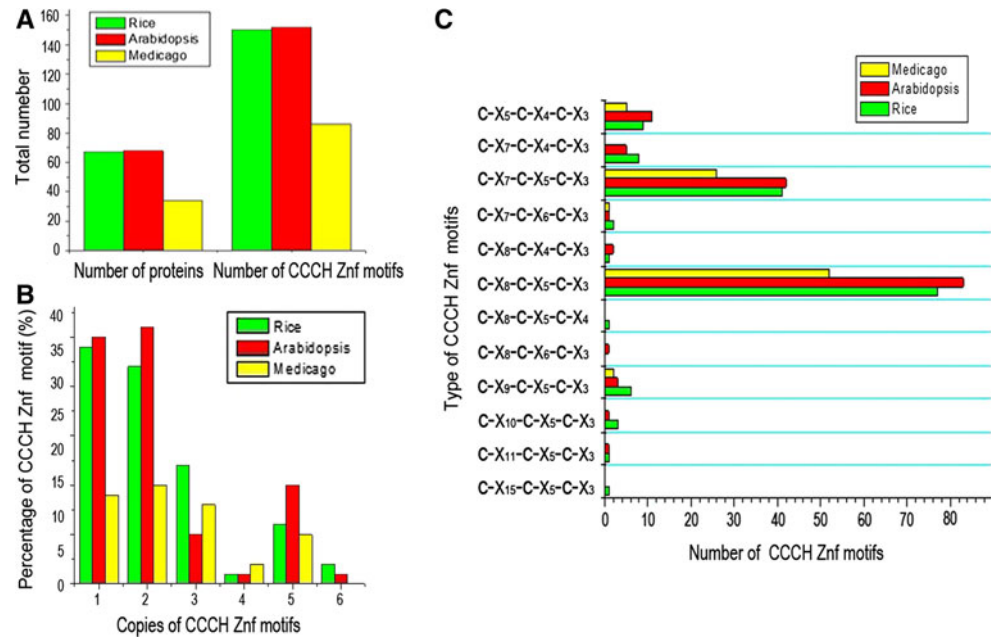
Table 1 Characterization of the predicted *CCCH* Znf proteins found in *M. truncatula*

Gene name	Sequence ID	Number of amino acids	Molecular weight (kDa)	PI	Chr	Number of CCCH motifs
MtC3H1	IMGAIAC235678_19.1	139	15,861.4	8.77	No	2
MtC3H2	IMGAIAC235753_22.1	511	56,845.0	5.01	No	5
MtC3H3	IMGAIMedtr1g035130.1	272	30,749.9	7.08	1	2
MtC3H4	IMGAIMedtr1g116940.1	340	36,436.2	8.57	1	2
MtC3H5	IMGAIMedtr2g019740.1	482	50,982.9	8.51	2	4
MtC3H6	IMGAIMedtr2g103290.1	181	20,367.9	9.30	2	2
MtC3H7	IMGAIMedtr3g028170.1	481	54,062.5	5.19	3	3
MtC3H8	IMGAIMedtr3g028190.1	420	47,095.0	5.77	3	4
MtC3H9	IMGAIMedtr3g050390.1	246	27,770.5	7.98	3	2
MtC3H10	IMGAIMedtr3g095960.1	1,084	122,178.5	7.28	3	3
MtC3H11	IMGAIMedtr3g103960.1	278	31,838.9	8.66	3	2
MtC3H12	IMGAIMedtr4g043660.1	682	75,448.2	5.73	4	1
MtC3H13	IMGAIMedtr4g059550.1	850	93,034.2	5.75	4	1
MtC3H14	IMGAIMedtr4g078660.1	317	35,857.3	8.75	4	2
MtC3H15	IMGAIMedtr4g104260.1	573	63,117.7	4.73	4	5
MtC3H16	IMGAIMedtr4g113860.1	384	43,812.8	9.49	4	3
MtC3H17	IMGAIMedtr4g113890.1	231	25,861.6	8.62	4	3
MtC3H18	IMGAIMedtr4g113940.1	270	30,545.6	8.68	4	3
MtC3H19	IMGAIMedtr4g113970.1	297	33,995.5	8.85	4	3
MtC3H20	IMGAIMedtr4g119280.1	582	61,444.0	8.76	4	5
MtC3H21	IMGAIMedtr5g008900.1	685	74,926.2	5.81	5	1
MtC3H22	IMGAIMedtr5g070100.1	393	46,264.2	8.85	5	1
MtC3H23	IMGAIMedtr5g071070.1	762	84,228.3	6.38	5	2
MtC3H24	IMGAIMedtr5g082830.1	668	73,480.6	6.07	5	2
MtC3H25	IMGAIMedtr5g085200.1	422	46,179.9	8.51	5	5
MtC3H26	IMGAIMedtr5g090040.1	159	18,810.9	9.79	5	1
MtC3H27	IMGAIMedtr7g079840.1	418	46,005.7	8.09	7	5
MtC3H28	IMGAIMedtr7g092070.1	734	80,438.1	6.04	7	2
MtC3H29	IMGAIMedtr7g093130.1	301	32,266.5	9.47	7	3
MtC3H30	IMGAIMedtr7g108860.1	394	42,699.2	4.66	7	1
MtC3H31	IMGAIMedtr7g109250.1	387	44,366.6	7.22	7	1
MtC3H32	IMGAIMedtr8g022240.1	396	45,004.0	9.82	8	1
MtC3H33	IMGAIMedtr8g058480.1	358	39,067.3	4.85	8	1
MtC3H34	IMGAIMedtr8g066730.1	376	39,887.7	6.56	8	3

rice (Fig. 1b). After whole-genome analysis of *CCCH* Znf proteins in *Arabidopsis* and rice, the *CCCH* Znf proteins were re-defined as C-X₄₋₁₅-C-X₄₋₆-C-X₃-H based on the different number of amino acid spacers between cysteines and histidines (Wang et al. 2008a), which were originally defined as C-X₆₋₁₄-C-X₄₋₅-C-X₃-H (Berg and Shi 1996). Similar to that reported in *Arabidopsis* and rice, the most abundant *CCCH* Znf motifs were found to be C-X₈-C-X₅-C-X₃-H (60 %) and C-X₇-C-X₅-C-X₃-H (30 %) in the Medicago *CCCH* Znf gene family (Fig. 1c). The detailed information of 34 Medicago *CCCH* Znf genes can be found in Table S2.

Besides containing one to five copies of *CCCH* Znf motifs, some *CCCH* Znf proteins also carry several other known functional domains, including ANK, KH, RRM, ZF-C3H4, SAP, CS and ZF-Ring (Fig. 2). It is worth mentioning that among the 34 *CCCH* Znf proteins, two proteins contain the KH domain, and four members contain the RRM domain. KH and RRM are common RNA-binding domains in eukaryotes (Lorkovic and Barta 2002; Lunde et al. 2007; Wei et al. 2007). Moreover, KH and RRM domains containing proteins have been proved to play essential roles in many aspects of RNA metabolism, suggesting that the six *CCCH* Znf proteins containing these

Fig. 1 Characterizations of the *CCCH* Znf proteins. **a** Number of *CCCH* Znf proteins and *CCCH* Znf motifs identified in *Medicago*, rice and *Arabidopsis*. **b** Percentage of different copies of *CCCH* Znf motifs in *Medicago*, rice and *Arabidopsis*. **c** Number of different types of *CCCH* Znf motifs in *Medicago*, rice and *Arabidopsis*



domains may function as RNA-binding proteins and are involved in RNA processing. In addition, we noted that three members of subfamily VI contained the ankyrin (ANK) motif which was demonstrated to play a variety of roles in diverse molecular processes such as transcriptional initiation, ion transport and signal transduction (Becerra et al. 2004; Mosavi et al. 2002). These specific motifs may, to some extent, contribute to the functional divergence of *CCCH* Znf genes (Wang et al. 2008a).

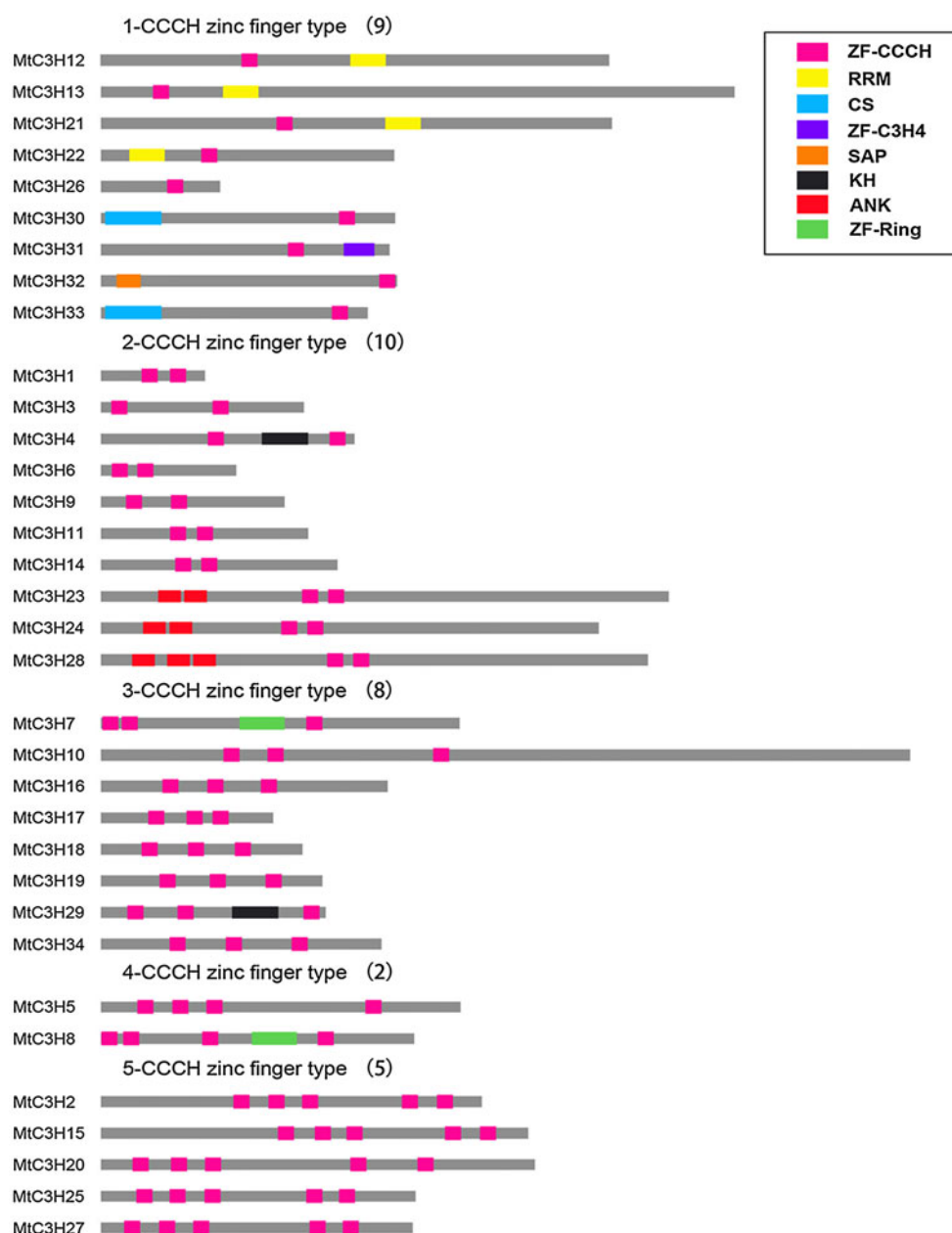
Phylogenetic and structural analysis of *CCCH* Znf proteins in *M. truncatula*

To evaluate the phylogenetic relationships within the *CCCH* Znf gene family in *M. truncatula*, a phylogenetic tree was generated based on the alignment of the full-length sequences of the 34 *Medicago CCCH* Znf proteins. According to the phylogenetic analysis, 34 *Medicago CCCH* Znf genes were divided into nine subfamilies (bootstrap values ≥ 500), designated I–IX. Thirty-four *Medicago CCCH* Znf genes formed 12 sister pairs, and 9 of them displayed high bootstrap support (≥ 998). We also noted that the majority of the phylogenetic clades had well-supported bootstrap values, but the bootstrap values of some proteins were low at the nodes and the phylogenetic relationships were unclear (Fig. 3). Therefore, the genes with low bootstrap value were not taken into consideration for further analysis. To complement the phylogenetic analysis, we executed an exon–intron analysis by comparing the predicted coding sequences with the genomic sequence of the 34 *Medicago CCCH* Znf genes (Fig. 3). Consistent with the phylogenetic analysis, we found that

members belonging to the same subfamily displayed similar exon–intron structures, especially in the number of introns, although exceptions to this observation were also found. For example, five *CCCH* Znf genes in subfamily I contain one intron except *MtC3H16*, which contain two introns. Interestingly, although the exon–intron organization of *CCCH* Znf genes varied considerably in terms of intron numbers, the intron phase was remarkably well conserved, indicative of exon shuffling during the evolution process (Kolkman and Stemmer 2001).

With the development of comparative genomics, it is possible to analyze the same gene families among different species. To investigate the evolutionary relationships of the *CCCH* Znf genes in *Medicago*, *Arabidopsis* and rice, a combined N–J tree was constructed from alignments of the complete sequences of all 169 *CCCH* Znf proteins from the three species. We divided the 169 members into 20 groups, designated from A to T (bootstrap support ≥ 700). The phylogenetic analysis displayed that most *Medicago CCCH* Znf genes were clustered with their rice or *Arabidopsis* counterparts with high bootstrap support (Fig. 4). Furthermore, we noted that the *MtC3Hs* were more closely grouped with the *AtC3Hs* than with the *OsC3Hs*. For example, ortholog pairs of *Medicago* and *Arabidopsis CCCH* Znf proteins were more prevalent in the tree, which implied that some ancestor *CCCH* Znf genes had been existing before the divergence of *Medicago* and *Arabidopsis*. Generally speaking, the dicot *CCCH* Znf genes formed sister pairs with the nearest dicot orthologs, while the monocot *CCCH* Znf genes formed sister pairs with the nearest dicot orthologs. However, sister pairs formed by rice and *Medicago/Arabidopsis* orthologs with high

Fig. 2 Schematic structures of CCCH Znf proteins in *M. truncatula*. Schematic structures of 34 CCCH Znf proteins are shown with names of all members on the left side of the figure. Different domains are indicated by different colored boxes denoted at the right top corner. The proteins were grouped manually according to the number of CCCH zinc fingers and the distribution of other conserved domains (colour figure online)



bootstrap values were found in the phylogenetic tree, such as *OsC3H45*–*AtC3H11* and *OsC3H62*–*MtC3H32*, while some groups in the tree consisting of a few CCCH Znf genes of Medicago, *Arabidopsis* and rice were significantly expanded, like group K. Previous study demonstrated that the *Arabidopsis* CCCH Znf genes clustered in group K in our phylogenetic tree are involved in responses to abiotic stress, suggesting that Medicago CCCH Znf genes of this group may play critical roles in plant stress responses, since genes clustered into the same group of phylogenetic tree often share similar functional features (Wang et al. 2008a). CCCH Znf genes with unclear phylogenetic relationships were also found in this tree.

Chromosomal location and gene duplication

Based on the starting position of each Medicago CCCH Znf genes on chromosomes, the 34 CCCH Znf genes were found to be unevenly distributed on seven chromosomes, except chromosome 6. Moreover, we noted that two CCCH Znf genes (*MtC3H1* and *MtC3H2*) were located on currently unattributed scaffold fragments, named Chr. 0 (Fig. 5). Chromosome 4 contained the largest number of CCCH Znf genes (nine). By contrast, only two CCCH Znf genes were located on chromosomes 1 and 2. Three CCCH Znf genes were identified on chromosome 8, while 15 CCCH Znf genes (5 each) were mapped to chromosomes 3, 5 and 7.

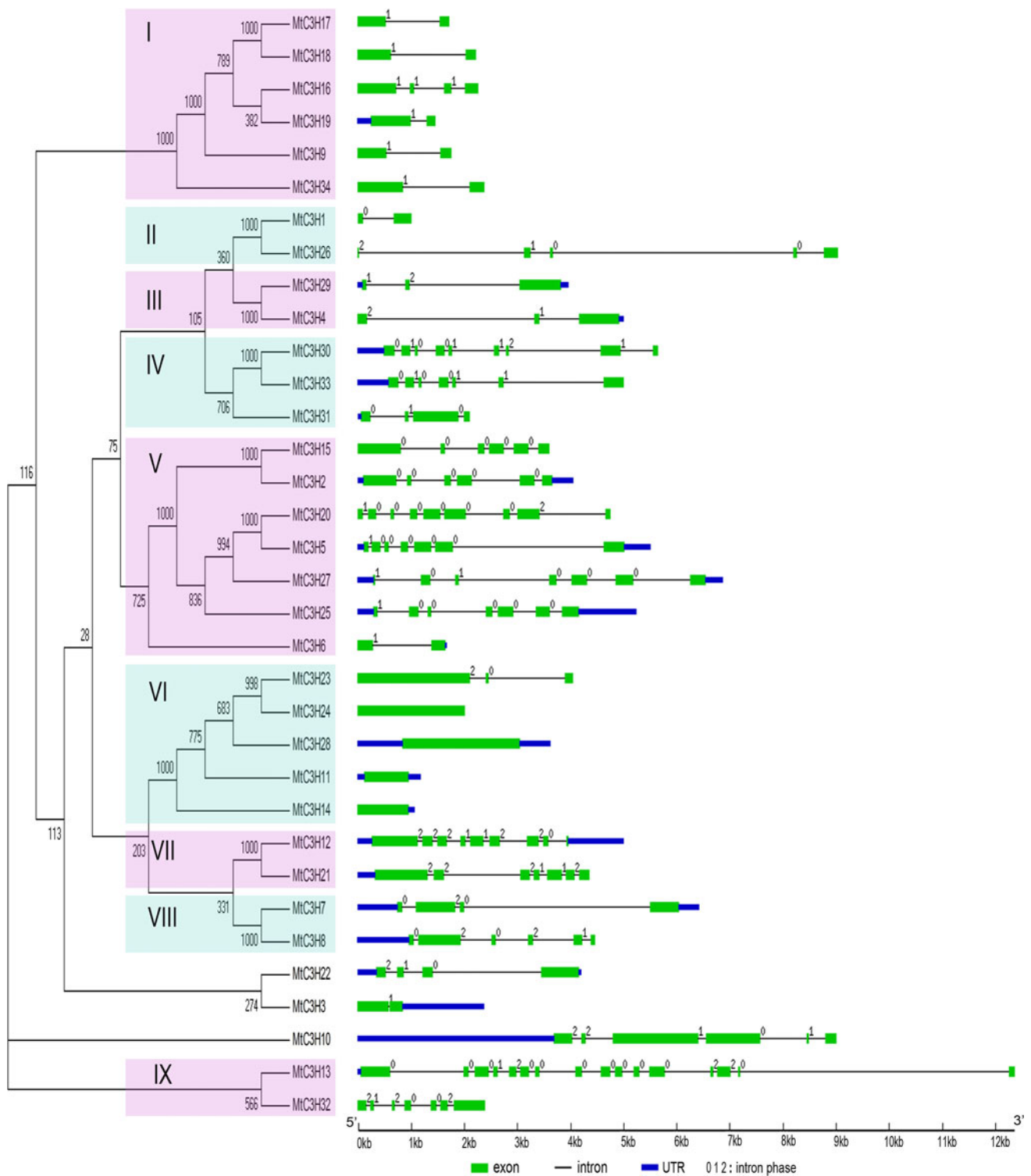


Fig. 3 Phylogenetic relationships and exon–intron structure of *Medicago* CCCH Znf proteins. The following parts are shown from left to right. Left an unrooted phylogenetic tree constructed using ClustalX (1.83) by the N-J method, summarizes the evolutionary relationship among the 34 members of the CCCH Znf family. The tree shows the nine major phylogenetic subfamilies (left column, numbered I–IX and marked with different alternating tones of blue

and red to clarify subfamily identification easier) with high predictive value. Right exon–intron structure; the exons and introns are indicated by green rectangles and thin lines, respectively. The untranslated regions (UTRs) are indicated by thick blue lines. Intron phases 0, 1, and 2 are indicated by the numbers 0, 1 and 2, respectively (colour figure online)



Fig. 4 Phylogenetic relationships of *Medicago*, rice and *Arabidopsis* *CCCH* Znf proteins. The tree was constructed based on the alignment of the full-length amino acid sequences of *Medicago*, rice and *Arabidopsis* *CCCH* Znf proteins by the neighbor-joining method using ClustalX. Bootstrap values (≥ 500) based on 1,000 replications are shown at each node

Based on the definition of gene clusters, a chromosomal region containing two or more genes within 200 kb can be considered to be a gene cluster (Holub 2001). In *M. truncatula*, only six *CCCH* Znf genes were identified in two clusters, located on chromosomes 3 and 4 (Fig. 5).

Previous studies have found that gene duplication and divergence events were the main contributors to evolutionary momentum (Bowers et al. 2003; Vision et al. 2000). Gene duplication events, including tandem and segmental duplications, were investigated with the purpose of explaining the mechanism of expansion of the *Medicago* *CCCH* Znf gene family during the evolutionary process. The definition of a gene duplication event is as follows: (1) the length of the aligned sequence covers $\geq 80\%$ of the longer gene; and (2) the similarity of the aligned regions is $\geq 70\%$ (Gu et al. 2002; Yang et al. 2008). In this study, three gene pairs participated in gene duplication, among which two pairs belonged to segmental duplication and the other one to tandem duplication (Fig. 5).

Expression profiles of *CCCH* Znf genes in *M. truncatula*

To investigate the expression patterns of the *Medicago* *CCCH* Znf genes, a comprehensive expression analysis was performed based on whole-genome coverage. We have

analyzed the *Medicago* microarray data generated by (Wilkins et al. 2009). Ten *Medicago* *CCCH* Znf genes (*MtCCCH1*, 2, 4, 9, 16, 17, 18, 26, 30 and 33) did not have their corresponding probe sets in the dataset, but the expression profiles of the rest of the 24 *CCCH* Znf genes were analyzed (Fig. 6). The majority of *CCCH* Znf genes showed distinct tissue-specific expression patterns across the seven tissues examined. For instance, three genes (*MtCCCH14*, 21, 32) showed the highest transcript abundances in vegetative buds, and *MtCCCH32* also showed high expression in the seed pod. *MtCCCH11* and *MtCCCH23* were both preferentially expressed in roots. *MtCCCH24* and *MtCCCH27* had the highest accumulation in stem and leaf, respectively. The results revealed that the *CCCH* Znf genes belong to even the same subfamily would be different in each tissue.

To gain insight into the stress responses of the subfamily VI, we analyzed their expression profiles under three stress conditions (PEG-6000, NaCl and ABA) using semi-quantitative RT-PCR (Fig. 7). All members of subfamily VI can be detected under normal growth conditions with the exception of *MtCCCH14*, which was also not expressed under the other three conditions. The results showed that the other four *Medicago* *CCCH* Znf genes of subfamily VI exhibited significant differences in their expression levels in response to one or more stress treatments. Three genes (*MtCCCH11*, 24 and 28) were significantly upregulated in response to ABA treatment, and following PEG-6000 treatment the expression levels of the four *Medicago* *CCCH* Znf genes (*MtCCCH11*, 23, 24 and 28) were obviously increased. But under NaCl treatment, these four genes did not show any significant changes in expression levels.

Fig. 5 Chromosomal locations of 34 *Medicago* *CCCH* Znf genes. The scale on the left is in megabases. The chromosome numbers are indicated at the top of each chromosome. The gene names are on the left side of each chromosome according to the approximate physical location of each *CCCH* gene. The unmapped *CCCH* Znf genes are shown on Chr. 0. The segmentally duplicated genes are connected by red lines, and the tandemly duplicated genes are marked in pink. Green line indicates a gene cluster (colour figure online)

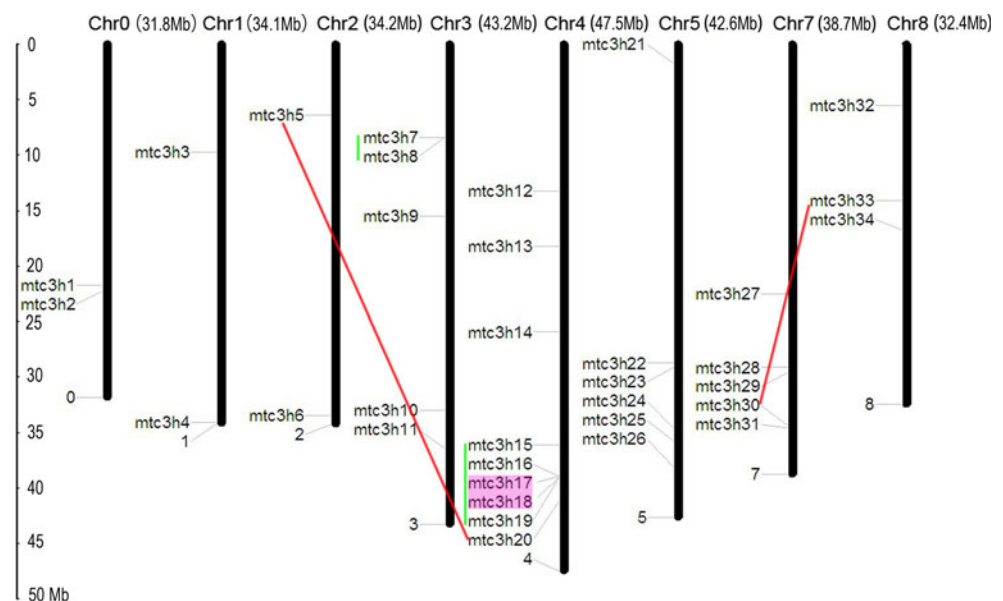
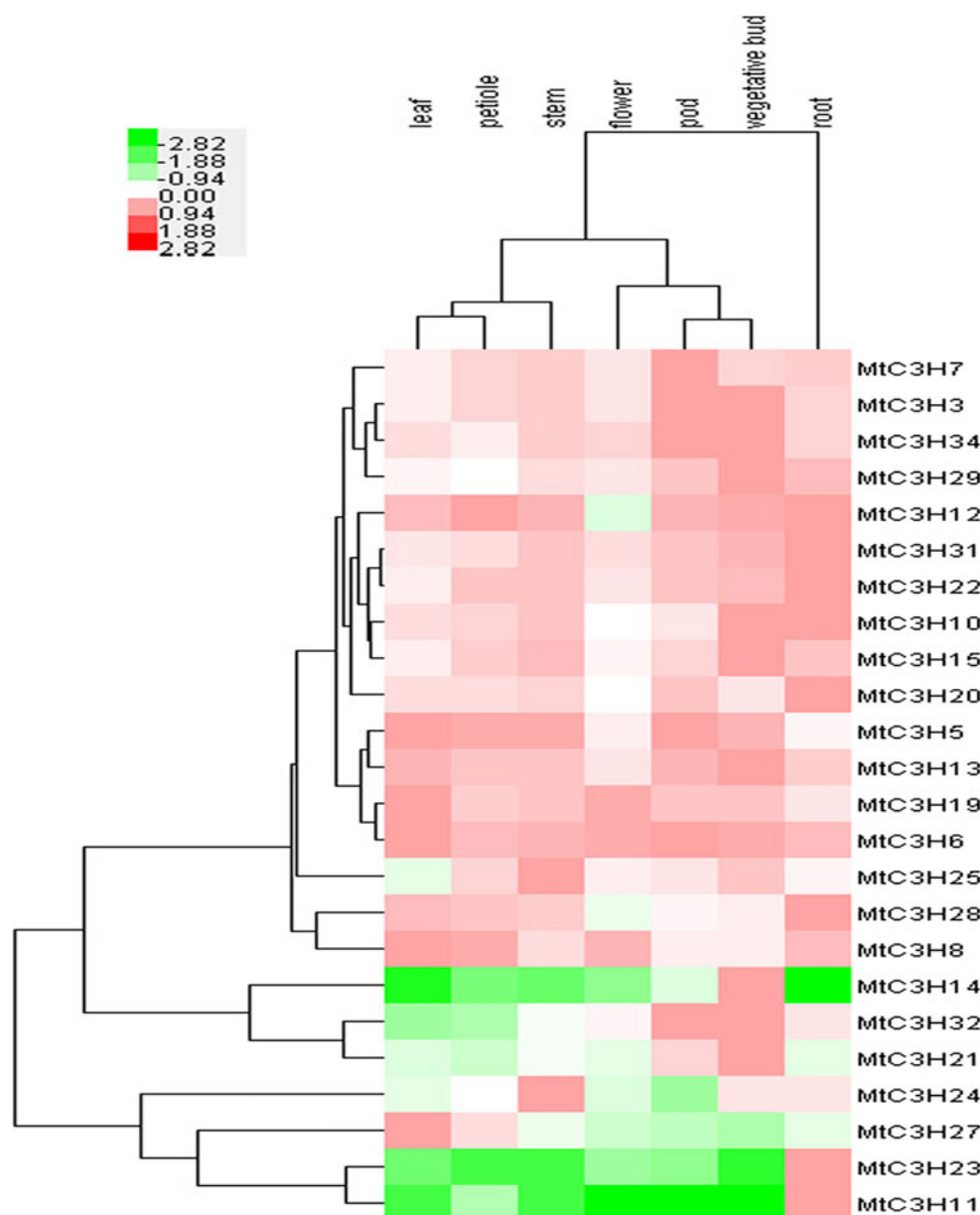


Fig. 6 Hierarchical clustering of *CCCH* Znf proteins in six tissues of *M. truncatula*. The different colors correspond to the log-transformed values of the protein change-fold ratio shown in the bar at the upper left of the figure. The color scale at the top of each dendrogram represents log₂ expression values, green represents a low level and pink indicates a high level of transcript abundance (colour figure online)



Discussion

Previous studies have shown that the *CCCH* Znf proteins play important roles in many aspects of plant growth and development. The features and functions of *CCCH* Znf gene family were surveyed and identified in *Poplar*, maize, *Arabidopsis* and rice (Chai et al. 2012; Peng et al. 2012; Wang et al. 2008a). Genome-wide analysis of the *CCCH* Znf genes in model legume plant *M. truncatula* would facilitate a better understanding of this gene family. In this study, we identified 34 *CCCH* Znf genes in the *M. truncatula* genome. As the results indicated, we noted that the number of *CCCH* Znf genes in *M. truncatula* was approximately half of that present in *Arabidopsis* (67) and rice (68),

and the number of *CCCH*-type Znf motifs in *M. truncatula* (5) was far less than in *Arabidopsis* (10) and rice (11) (Wang et al. 2008a). These may suggest that the number of *CCCH* Znf genes in the three species is not directly linked to genome size, since the *Medicago* genome size is approximately three times that of the *Arabidopsis* genome and similar in size to the rice genome (Khurana and Gaikwad 2005). However, it is worth noting that the five types of *CCCH* Znf motifs present in *Medicago* all exist in *Arabidopsis* and rice, and 90 % of the identified *Medicago* *CCCH* Znf motifs are C-X₇₋₈-C-X₅-C-X₃-H motifs, similar to those found in *Arabidopsis* (83 %) and rice (77 %). This trend implies that the *CCCH* Znf motifs in *Medicago* are highly conserved compared with *Arabidopsis* and rice *CCCH* Znf motifs and

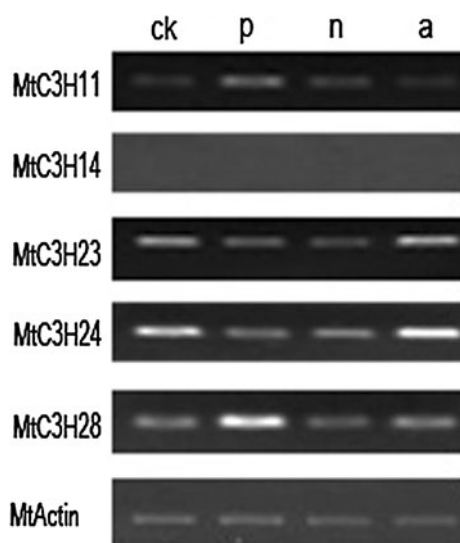


Fig. 7 Semi-quantitative RT-PCR analysis of subfamily VI genes under different stress conditions. *ck*, unstressed control; *p*, PEG-6000 (drought); *n*, NaCl (salt); *a*, abscisic acid. Stress treatments and time course are described in “Materials and methods”

also suggests that the C-X₇₋₈-C-X₅-C-X₃-H motifs play an important functional role in the *CCCH* Znf proteins.

In this study, the 34 *Medicago* *CCCH* Znf genes were classified into nine subfamilies based on phylogenetic analysis. Some genes were difficult to assign to any subfamilies because of their low bootstrap values (<500) in the clades. Interestingly, this phenomenon also exists in *Arabidopsis* and rice *CCCH* Znf gene families. However, sequence analysis revealed that the *CCCH* Znf motifs are highly conserved, while the other characteristics are diverse, especially the types of *CCCH* Znf motifs present in each protein and the spacing in protein sequences between adjacent *CCCH* Znf motifs. It is notable that the *CCCH* Znf genes belonging to the same subfamily always display a similar domain architecture, suggesting that these *CCCH* Znf genes may have similar functions. A combined N-J tree was also constructed to investigate the phylogenetic relationships of *CCCH* Znf genes in the three representative species and their evolutionary relationships. From the phylogenetic tree, we noted that some ortholog genes showed closer relationships than the paralog genes among *Medicago* and *Arabidopsis* *CCCH* Znf genes, suggesting that the ortholog genes may have originated from a common ancestor when duplication events occurred. Additionally, the ortholog genes from the monocot (rice) and dicot (*Medicago* and *Arabidopsis*) were also found in the phylogenetic tree, implying that these ortholog genes originated from common ancestral genes which existed before the divergence of monocots and dicots. These observations suggested that the ortholog genes were highly conserved during the evolutionary process.

Gene duplication can be divided into tandem and segmental duplication events and plays an important role in a succession of genomic rearrangements and expansions (Vision et al. 2000). Previous studies have shown that segmental duplication is largely responsible for the expansion of *Arabidopsis* and rice *CCCH* Znf gene families (Wang et al. 2008a). In this study, three gene pairs (two segmental duplication and one tandem duplication) participated in gene duplication, which is far less than that found in *Arabidopsis* and rice. It may be related to the number of *CCCH* Znf genes present in the three species. However, the proportion of segmental duplication of *CCCH* Znf gene pairs in *Medicago* is consistent with that in *Arabidopsis* and rice (Wang et al. 2008a). These results suggest that segmental duplication is also the main contributor to the expansion of the *CCCH* Znf gene family in *Medicago*. It is believed that segmental duplication events occur more often in the more slowly evolving gene families (Cannon et al. 2004). The prevalence of ortholog pairs and segmentally duplicated genes in the three different species suggests that the *CCCH* Znf gene family is a conserved and slowly evolving family in plant genomes. This suggestion also gives us a possible reason for the similar trends of *CCCH* Znf proteins present in the three species.

Recent studies have demonstrated that the *CCCH* Znf gene family in plants was involved in stress response in plants (Chai et al. 2012; Peng et al. 2012; Wang et al. 2008a). In this study, all members of subfamily VI were predicted to be involved in response to stress treatments based on phylogenetic analysis. Through semi-quantitative RT-PCR analysis, we observed that four *Medicago* *CCCH* Znf genes had significantly higher or lower expression levels, when the seedlings were exposed to different stresses (PEG-6000, NaCl and ABA). However, *MtC3H14* was not detected in any stress condition, even in normal growth condition. It may due to structural imperfections in the protein. Although the five genes belong to the same subfamily and have close evolutionary relationships, their expression profiles are different. The different expression profiles suggest that these *CCCH* Znf genes carry out different physiological functions in the host adaptation to complicated environmental challenges. Notably, *MtC3H24* and *MtC3H28* were strongly induced by ABA and PEG-6000 treatments, respectively, strongly indicating that they may play essential roles in response to abiotic stress.

M. truncatula is an important leguminous model plant that is widely used in physiological, genetic and molecular studies of legumes. The present study has identified and characterized the *CCCH* Znf genes in *M. truncatula*. This should provide a solid foundation for future functional studies and also in guiding future experimental work on *CCCH* Znf genes in this model species.

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